

A Comparison of Optimizers for Text Classification

Introduction

This paper studies machine learning models for text classification. The introduction provides background and motivation for comparing optimization strategies under a fixed compute budget.

Our motivation is to understand how gradient descent variants affect final model accuracy on a held-out evaluation set.

Methodology

We train each model with stochastic gradient descent and report the mean accuracy across five random seeds. The methodology section describes the learning rate schedule and batch size.

Table 1: Accuracy comparison across optimizers

Optimizer	Accuracy	F1	Epochs
SGD	0.842	0.831	20
Adam	0.910	0.905	20
AdamW	0.921	0.918	20

Results

The results show that AdamW achieves the highest accuracy and F1 performance in our comparison. Gradient descent with momentum trailed the adaptive optimizers.

Table 2: Performance comparison by dataset

Dataset	Accuracy	Latency_ms
News	0.918	12
Reviews	0.904	11